

# Review of Probability and Statistics

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# Roadmap

Bivariate Probability

Covariance and Correlation

So far, we've only looked at a single random variable  $Y$

Now we turn to studying relationships between two random variables  $X$  and  $Y$

We are moving from univariate analysis to bivariate analysis

(Soon enough we will also do multivariate analysis)

We assume that we have available a random sample of observations on ordered pairs  $(X_i, Y_i)$  from an unknown population

We don't know the *marginal* population distributions of  $X_i$  and  $Y_i$

We also don't know the *joint* population distributions of  $X_i$  and  $Y_i$

Our focus in this section is to study parameters of the joint distribution

(We could, of course, use our knowledge from the previous two weeks to study  $X_i$  and  $Y_i$  separately; for example, we could calculate sample averages of both of them)

The easiest thing to do when studying the joint distribution is to take a look at the scatterplot

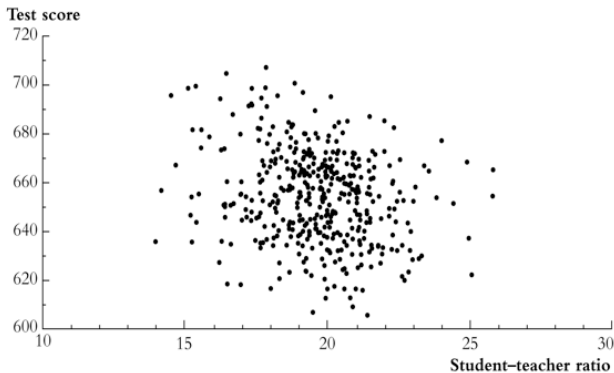
Let's say we are interested in looking at the joint distribution of test scores and student-teacher ratios

The primitive idea is that schools with lower student-teacher ratios achieve better results in standardized test scores

Using data from Californian school districts, the scatterplot looks like this...

**FIGURE 4.2** Scatterplot of Test Score vs. Student-Teacher Ratio (California School District Data)

Data from 420 California school districts. There is a weak negative relationship between the student-teacher ratio and test scores: The sample correlation is  $-0.23$ .



In the univariate world, the main parameters of interest were the population mean and the population variance

We have learned that the sample average is the best estimator of the population mean

In the bivariate world, the main parameters of interest are the population covariance and the population correlation

We will now define the population parameters and their sample analogs

### Definition

The **population covariance** between  $X$  and  $Y$  is defined by

$$\text{Cov}(X, Y) := E[(X - E[X])(Y - E[Y])]$$

The population covariance is sometimes also denoted by  $\sigma_{XY}$ .

### Definition

The **sample covariance** between  $X$  and  $Y$  is defined by

$$s_{XY} := \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$



### Definition

The **population correlation** between  $X$  and  $Y$  is defined by

$$\rho_{XY} := \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}$$

### Definition

The **sample correlation** between  $X_i$  and  $Y_i$  is defined by

$$r_{XY} := \frac{s_{XY}}{s_X \cdot s_Y}$$

Do you remember what  $s_X$  and  $s_Y$  are?

Covariance and correlation measure joint variation in  $X$  and  $Y$

If above average values of  $X$  tend to occur together with above average values of  $Y$ , then both covariance and correlation will be positive

If below average values of  $X$  tend to occur together with below average values of  $Y$ , then both covariance and correlation will be positive (not a typo)

If above average values of  $X$  tend to occur together with below average values of  $Y$ , then both covariance and correlation will be negative

If below average values of  $X$  tend to occur together with above average values of  $Y$ , then both covariance and correlation will be negative

What's the difference between covariance and correlation?

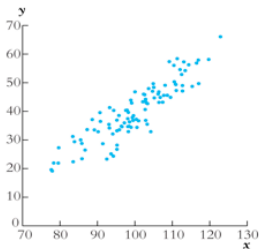
Covariance inherits the units of measurement of both  $X$  and  $Y$ , in fact it multiplies them

Hard to interpret what the unit of measurement of the product of  $X$  and  $Y$  are

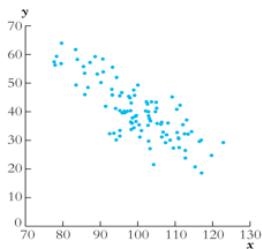
Correlation, by dividing the covariance by the standard deviations, removes the units of measurement and instead results in a unit-free number that has to lie between -1 and 1

In that sense, the correlation is easier to interpret

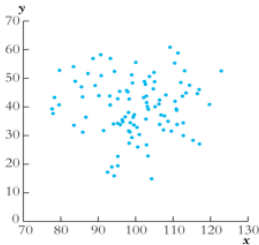
Limitation: covariance and correlation only capture linear relationship between  $X$  and  $Y$



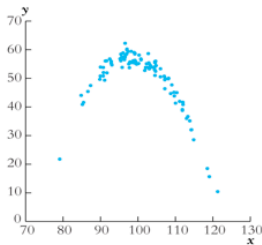
(a) Correlation = +0.9



(b) Correlation = -0.8



(c) Correlation = 0.0



(d) Correlation = 0.0 (quadratic)